

Incident Investigation: Deepwater Horizon Macondo Well Blowout (2010)

1. Executive Summary

On April 20, 2010, the Deepwater Horizon drilling rig operating in Mississippi Canyon Block 252 experienced a catastrophic blowout, explosion, and fire. The incident resulted from a failure of primary well control barriers followed by unsuccessful actuation of the subsea Blowout Preventer (BOP) stack.

2. Well Control Barrier Breakdown

Investigation revealed that the nitrogen-foamed cement slurry pumped at the shoe track of the 9-7/8 inch production casing failed to isolate hydrocarbons in the pay zone. Hydrocarbons entered the wellbore undetected until gas expanded above the subsea wellhead.

3. Negative Pressure Test Anomalies

A critical negative pressure test conducted prior to temporary abandonment exhibited 1,400 psi on the drill pipe while zero pressure was measured on the kill line. Rig personnel misinterpreted this differential pressure as a 'bladder effect' or thermal artifact rather than an ongoing hydrocarbon kick.

4. Blowout Preventer (BOP) Failure Analysis

When hydrocarbons breached the surface, the annular preventers were closed, but high fluid velocity eroded the packing element. Subsequent activation of the blind shear rams (BSR) failed to seal the well because the drill pipe had buckled off-center under extreme differential pressure and mechanical compression.